Data and Features

Getting data ready for a model to learn from.



Today's Objective

Transforming raw, messy real-world data into structured numerical inputs that machine learning algorithms can analyze and learn from.

AGENDA

Today's Plan: From Messy Data to Numbers

LAST WEEK IN ONE LINE: A model is a pile of numbers, tuned by walking downhill on its mistakes. But where do input numbers come from?

1

Know Your Data

Data types, structured tables, feature scaling, and handling missing values effectively.

2

Feature Extraction

Converting unstructured formats like text, photos, and sound into quantitative vectors.

3

Similarity

Mathematical measurements of closeness, Euclidean distance, and cosine similarity.

4

Keep, Drop, Build

Selecting valuable columns, eliminating data leakage, and engineering new features.

KNOW YOUR DATA

Six Kinds of Data You Will Meet

Numbers

Age, price, temperature, counts

Categories

Colour, country, yes / no options

Text

Reviews, messages, emails, documents

Images

Photos, scans, medical X-rays

Audio

Voice notes, music clips, alarms

Time Series

Daily sales, heartbeats, stock prices

Core Principle: Models only understand numbers. Every kind above must be converted first.

KNOW YOUR DATA

The Anatomy of a Dataset

Student (ID)	Study hours (Feature)	Attendance % (Feature)	In a club? (Feature)	Passed? (Label)
S01	6	90%	yes	yes
S02	2	60%	no	no
S03	4	75%	no	yes
S04	1	50%	yes	no

Row

One individual example or observation (e.g., one student).

Features

What we know about the example (hours, attendance, club).

Label

What we want to predict or target (passed?).

Note: The student ID is just an identifier tag. It helps us locate a row but contains no predictive power.

Turning Categories into Numbers

ORDERED CATEGORIES

Use Sequential Integer Encoding

T-shirt size has a natural sequence and order. Numbers preserve this relative magnitude.

```
small  → 1
medium → 2
large  → 3
```

UNORDERED CATEGORIES

One-Hot Encoding

Colours have no intrinsic order. Create individual binary indicator columns for each value.

colour	is_red	is_blue	is_green
red	1	0	0
blue	0	1	0
green	0	0	1

⚠ Common Trap: Never label unordered categories 1, 2, 3! The model will assume green (3) is mathematically "more" than red (1).

Scaling: Give Every Feature a Fair Voice

Before Scaling

```
age: 30
salary: 60,000
```

Large numerical ranges dominate distance calculations. Salary completely drowns out age impact.

After Min-Max Scaling (0–1)

```
age: 0.24
salary: 0.40
```

Features are normalized onto a uniform scale, giving every variable an equal opportunity to contribute.

Min-Max Normalization Formula:

```
scaled = (value - min) ÷ (max - min)
```

Example: Ages 18 to 68 → (30 – 18) ÷ 50 = **0.24** | Salaries 20k to 120k → (60k – 20k) ÷ 100k = **0.40**

Missing Values: Handling Holes in Data

Student	Hours	Attendance	Passed
S01	6	90%	yes
S02	2	?	no
S03	4	75%	yes

1. Drop the Row

Effective when missingness is rare and abundant total training data is available.

2. Impute / Fill in

Replace missing entries with the column average, median, or most frequent category.

3. Add a Flag Column

Create an explicit indicator column like `was_missing?`. Missingness itself can be predictive.

Practical Rule of Thumb: If the majority of a column is empty, drop the column instead of dropping the majority of your dataset's rows.

FEATURE EXTRACTION

Measuring Raw Media to Extract Features

Text Input

`"Great food, slow service!"`

➔ Word counts, total length (4 words), contains exclamation mark (yes).

Photo Input

`A photo of a banana`

➔ Yellow color score, curvature measure, overall brightness.

Audio Clip

`5-second voice recording`

➔ Average loudness, fundamental pitch, speaking rate.

Date Value

`"Sat 12 Oct 2026"`

➔ Day of week (6), month (10), is_weekend (1).

Feature Extraction = Deciding what metrics to quantify so numerical arrays retain original semantic context.

FEATURE EXTRACTION

Text to Numbers: Bag of Words Model

Review 1:
`"good food good service"`

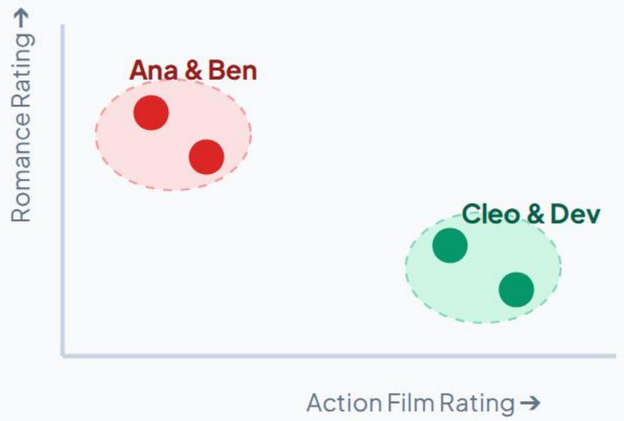
Review 2:
`"bad service"`

Document / Vector	Count of "good"	Count of "bad"	Count of "food"	Count of "service"
Review 1	2	0	1	1
Review 2	0	1	0	1

Representation: Review 1 becomes `[2, 0, 1, 1]` and Review 2 becomes `[0, 1, 0, 1]`.
Catch: Sequential word order is discarded (like shaking words in a bag). Modern NLP models address this syntax loss.

SIMILARITY

Similar Things Sit Close Together



Feature Space Mapping

By rating preference across film genres, each user becomes a coordinate point on a vector spatial map.

Recommendation Engine Logic

If Ana loved a film, nearby point Ben likely will too. Distant point Cleo likely will not.

Application: E-commerce, music streaming, and video platforms locate mathematical neighbors to suggest content.

SIMILARITY

Measuring Closeness with Distance

Euclidean Distance Formula

Straight-line "ruler" distance calculation between points:

Ana (5, 1) vs Ben (4, 2):

1. Calculate differences: $5 - 4 = \mathbf{1}$, $1 - 2 = \mathbf{-1}$
2. Square differences: $1^2 = \mathbf{1}$, $(-1)^2 = \mathbf{1}$
3. Sum squared values: $1 + 1 = \mathbf{2}$
4. Square root sum: $\sqrt{2} \approx \mathbf{1.41}$

Distance to Ana (5, 1):

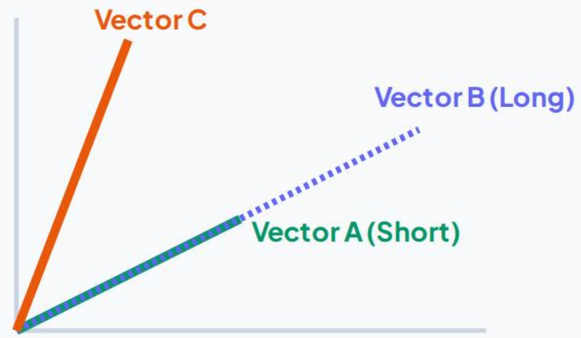
Ben (4, 2): 1.41 (Much Closer)

Distance to Ana (5, 1):

Cleo (1, 5): 5.66 (Further Away)

Smaller distance = higher similarity. Scales naturally from 2 features to 2,000+ features.

Cosine Similarity: Angular Orientation



Measures Angle, Ignores Length

Focuses exclusively on vector direction rather than absolute magnitude.

- **A & B: Same direction →** Cosine = **1.00**
- **A & C: 45° separation →** Cosine = **0.71**
- Scale: **1** (identical), **0** (unrelated), **−1** (opposite)

Text Match Advantage: A short review and long review discussing identical topics point in the same direction regardless of length.

SIMILARITY BETWEEN COLUMNS

Correlation: Do Two Columns Move Together?

Positive (+0.9)

Both columns rise together.
e.g., Study hours vs exam marks.

None (~0.0)

No observable pattern.
e.g., Shoe size vs exam marks.

Negative (−0.9)

One rises while other falls.
e.g., Phone price vs age of phone.

Redundancy Rule: Two columns with correlation near +1 or −1 convey duplicate information. Retain just one.

KEEP OR DROP

Six Reasons to Drop a Column

1. Pure ID

Student ID, phone numbers. Unique per row with no generalizable pattern to learn.

2. Mostly Empty

90%+ missing. Imputing values would mean artificially fabricating most data points.

3. Never Changes

Zero variance. Identical static value across all rows (e.g., "country = Ethiopia").

4. Duplicate

Repeats another column's signal (e.g., height in cm vs height in inches).

5. Leaky

Reveals future outcomes or information available only after event occurs.

6. Unfair / Private

Ethically inappropriate features for automated decision-making (e.g., protected traits).

Evaluation Check: Will this feature be known at prediction time? Does it provide distinct new information?

KEEP OR DROP

Data Leakage: The Column That Cheats

Known BEFORE Exam (Safe)

✔ Study hours

✔ Attendance %

✔ Past quiz scores

Known ONLY AFTER Exam (Leaky)

✘ Final grade letter

✘ Certificate issued?

✘ Re-sit booked?

The Leakage Trap: A model trained with leaky columns scores 100% in offline testing but fails completely in deployment because those features do not exist yet when making live predictions.

Building Better Features & The Golden Rule

Feature Engineering Transformations

COMBINE: price & size → price ÷ size = *price per m²*

SPLIT: "Sat 12 Oct" → *weekday, month, is_weekend*

GROUP: age 37 → *age band (30 to 45)*

COUNT: purchases list → *total monthly transactions*

THE GOLDEN RULE

Garbage In, Garbage Out.

Steps 1–3 (collecting, cleaning, and engineering data) take up most of the development timeline in real-world AI projects.

Recap in One Line:
Good models start with good, honest, well-prepared numerical data.